

Xenon

54
Xe

Like most members of its group, xenon is also an unreactive, colorless, and odorless element. However, it glows a striking blue when a high voltage is applied to it, and this property makes it a valuable component in car headlights and studio lights. Some rocket engines use xenon-based thrusters to propel the spacecraft forward.



Pure xenon electrified in a glass sphere



Lights on

Film studios use

xenon-based lamps, as they are very bright and produce light similar to natural sunlight. Special effects artists use this type of light source to create an intense, straight beam of light on film sets.

The Dawn spacecraft carried 937 lb (425 kg) of xenon at launch.



Rocket fuel

Xenon is used as a propellant in some rocket engines. Chemically inert and easily stored in a compact form, its atoms are heavy, so they provide more thrust compared to other propellants.

Artist's impression of NASA's Dawn spacecraft

Radon

86
Rn

Produced by the breakdown of uranium and other radioactive elements, radon is the only natural radioactive noble gas. In most places, this element is only present in tiny amounts, yet it is still responsible for most of the radiation on Earth. As radon moves up through the ground it can accumulate in buildings. This can become harmful, so it's important that its levels are checked and monitored.



Radioactive materials in this mineral break down to emit radon gas.

Uraninite

Glass sphere containing radon and air

The water from this volcanic hot spring contains radon.



Hot spring baths

Radon escapes from volcanic hot springs and muds along with other gases, such as carbon dioxide and sulfur dioxide. While radon is considered to be toxic by environmental agencies, some health-seekers visit thermal baths for a dip in these waters.